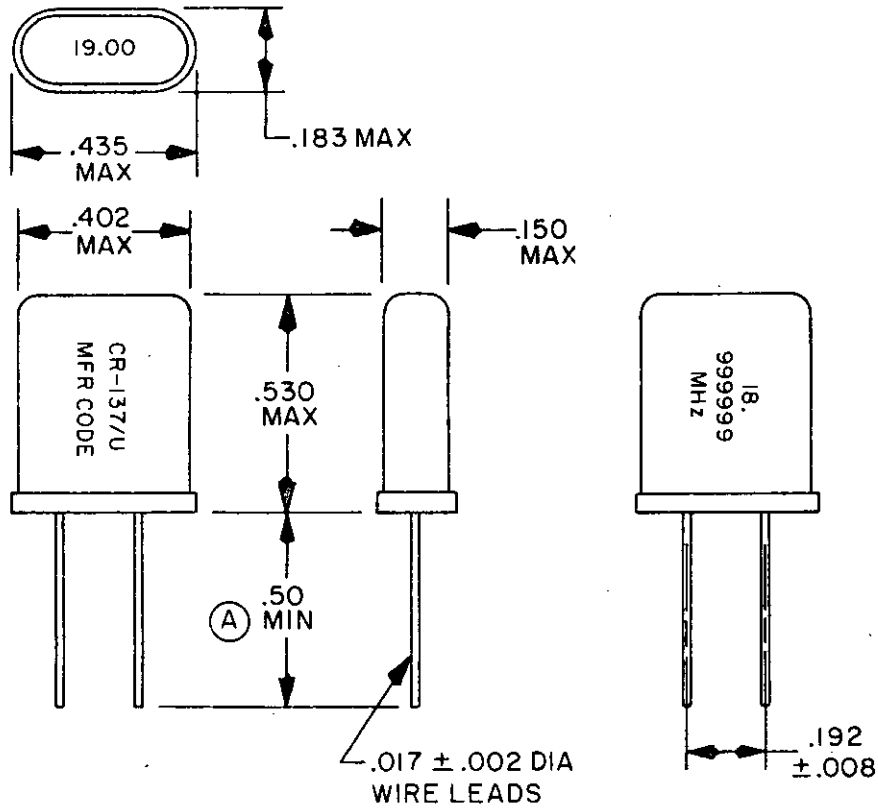
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-137/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 8.5 to 19.0 MHz; fundamental; noncontrolled; antiresonance.



INCHES	MM
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designator, frequency, and manufacturer's code shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Crystal unit - CR-137/U.

Ⓐ denotes changes

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FSC 5955

Ⓐ TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-18/U or HC-43/U (MIL-H-10056/9)
Dimensions, marking, and configuration	See figure 1.
Frequency range	8.5 to 19.0 MHz, inclusive
Capacitance, shunt	7.0 pF, maximum
Mode of oscillation	Fundamental
Test set, calibration values, and rated drive level	See table II
Antiresonance, load capacitance	32.0 ±0.5 pF
Reference temperature	40°C ±2°C
Operating temperature range	20°C to 70°C, inclusive
Frequency tolerances:	
Reference temperature	±25 ppm
Operating temperature range	±25 ppm of frequency measured at reference temperature
Equivalent resistance	See table III
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±5 ppm

1/ Method 201 of MIL-STD-202.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	MHz	Ohms	mA	Volt	mW
TS-330/TSM	8.5 to 15.0	25	10	----	2.5 ± 0.5
TS-683/TSM	15.0 + to 19.0	25	--	0.25	2.5 ± 0.5

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TABLE III. Equivalent resistance.

Frequency range inclusive	Maximum resistance
MHz	Ohms
8.5 to 9.999999	35
10.0 to 19.0	25

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11
DLA - ES

User activities:
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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